Dataset shift in neural models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable shift investigation record.

Retain group definitions, counts, model versions, missing labels and evaluation boundaries.

Keep one habit: Recall → W1–W3/W5 → independent mixture → project monitoring record → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A neural image classifier developed using mostly clear pictures may later receive mostly dim pictures. Its learned weights can stay unchanged while its overall results change.

  Dataset shift concerns changes in the data encountered, including **input**s, label frequencies or the relationship between **input**s and labels.

  A shift does not always reduce accuracy, and an accuracy change alone does not identify its cause.

- Use the fictional audit below. Clear and dim are disjoint groups defined before the audit, and every case belongs to exactly one group. The same model checkpoint, pre**process**ing and **prediction** threshold are used throughout.

  Counts are constructed teaching data, not results from a deployed model. Both cohorts have checked reference labels.

- Clear accuracy is 72 / 80 = 90 percent in development and 18 / 20 = 90 percent in new use. Dim accuracy is 10 / 20 = 50 percent and 40 / 80 = 50 percent.

  The observed group rates are unchanged, but the group proportions are different.

Uncertainty and failure analysis

> **My remembered idea, example and correction**

Similarity and normalisation

> **My remembered idea, example and correction**

Augmentation and leakage

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Cohort	Group	Cases	Correct
Development audit	Clear	80	72
Development audit	Dim	20	10
New-use audit	Clear	20	18
New-use audit	Dim	80	40

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed disjoint groups, unchanged model/preprocessing: development clear 80 cases/72 correct, dim 20/10; new use clear 20/18, dim 80/40. Clear rate 90% and dim 50% in both cohorts. Development correct 82/100, errors 18; new correct 58/100, errors 42. Drop 24 percentage points. Weighted development .8×.9 + .2×.5 = .82; new .2×.9 + .8×.5 = .58.

  Equal-sized 50/50 under these exact assumed rates gives 45+25 = 70 correct of 100. Separate picture: old bright 40/32, dim 10/5 gives 37/50 = 74%; new bright 10/8, dim 40/20 gives 28/50 = 56%. Bright rate is 80%, dim 50%; restoring the old picture mix restores 74% under the same rates. Do not mix the two examples. Without checked new labels, input mix can be monitored but current accuracy is unavailable.

  Check versions and label definitions; use new source-group-separated comparisons, preserve the original model and reserve final evaluation after changes. No causal lighting effect is established.

- From the table, calculate clear and dim accuracy in each cohort and the clear/dim proportions. Which observed quantities change, and which remain the same?

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main constructed disjoint groups, unchanged model/preprocessing: development clear 80 cases/72 correct, dim 20/10; new use clear 20/18, dim 80/40. Clear rate 90% and dim 50% in both cohorts. Development correct 82/100, errors 18; new correct 58/100, errors 42. Drop 24 percentage points. Weighted development .8×.9 + .2×.5 = .82; new .2×.9 + .8×.5 = .58.

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  Check versions and label definitions; use new source-group-separated comparisons, preserve the original model and reserve final evaluation after changes. No causal lighting effect is established.

- Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed disjoint groups, unchanged model/preprocessing: development clear 80 cases/72 correct, dim 20/10; new use clear 20/18, dim 80/40. Clear rate 90% and dim 50% in both cohorts. Development correct 82/100, errors 18; new correct 58/100, errors 42. Drop 24 percentage points. Weighted development .8×.9 + .2×.5 = .82; new .2×.9 + .8×.5 = .58.

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  Check versions and label definitions; use new source-group-separated comparisons, preserve the original model and reserve final evaluation after changes. No causal lighting effect is established.

- New-use **monitoring** shows more dim images but reference labels have not arrived. What can you report now, what can you not calculate, and what checks should precede blaming the neural network?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main constructed disjoint groups, unchanged model/preprocessing: development clear 80 cases/72 correct, dim 20/10; new use clear 20/18, dim 80/40. Clear rate 90% and dim 50% in both cohorts. Development correct 82/100, errors 18; new correct 58/100, errors 42. Drop 24 percentage points. Weighted development .8×.9 + .2×.5 = .82; new .2×.9 + .8×.5 = .58.

  Equal-sized 50/50 under these exact assumed rates gives 45+25 = 70 correct of 100. Separate picture: old bright 40/32, dim 10/5 gives 37/50 = 74%; new bright 10/8, dim 40/20 gives 28/50 = 56%. Bright rate is 80%, dim 50%; restoring the old picture mix restores 74% under the same rates. Do not mix the two examples. Without checked new labels, input mix can be monitored but current accuracy is unavailable.

  Check versions and label definitions; use new source-group-separated comparisons, preserve the original model and reserve final evaluation after changes. No causal lighting effect is established.

- A team retrains using new-use errors, then evaluates on those same images and their brightened copies. Explain why the score is not independent **evidence** and propose a comparison that includes the original model.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

  Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Analyse a new mixture independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **dataset shift** in neural models **evidence** sheet to Image-pattern Network Investigation. Show how you can explain how a changed **input** mix can change overall accuracy.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Use the right denominator for **Dataset shift** in neural models. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Dataset shift** in neural models. Proposed explanation: "Average the percentages regardless of group sizes". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

  Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Separate within-setting performance from changing setting proportions.

- Toy old-use mix: bright 40 cases 32 correct; dim 10 cases 5 correct. New-use mix: bright 10 cases 8 correct; dim 40 cases 20 correct.

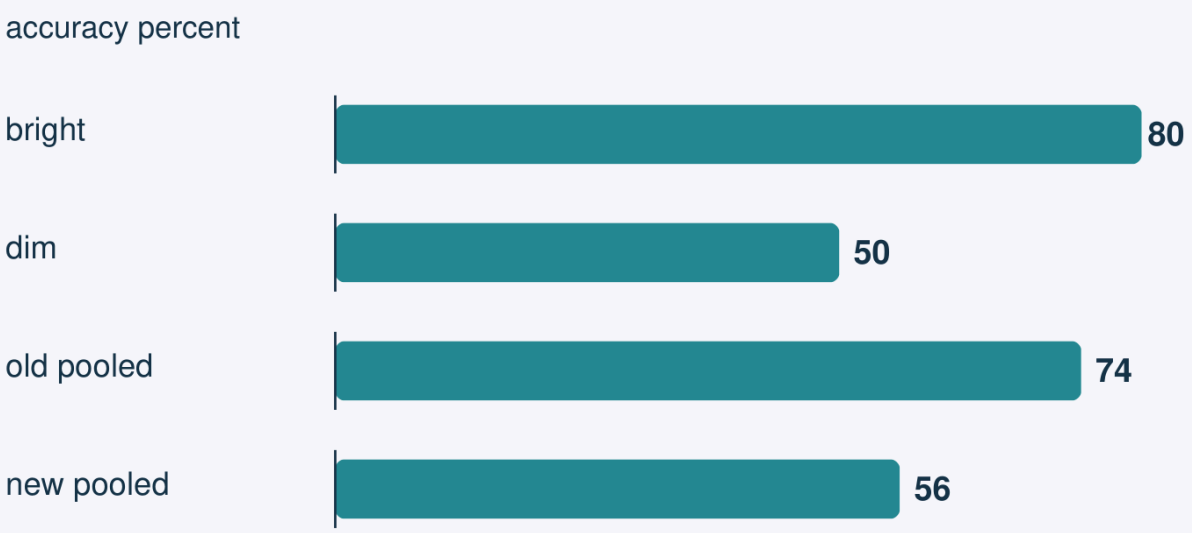
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Did the supplied model’s dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/30

Fixed slice rates, changed pooled accuracy



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/30

Dataset shift in neural models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep slice rates

Bright is 80% in both mixes; dim is 50% in both.



2. Old pooled rate

Old accuracy 37/50=74%.



3. New pooled rate

New accuracy 28/50=56%; more use now occurs in the harder dim setting.

CHANGE ONE THING

New-use mix returns to40 bright and 10 dim with the same slice rates.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate within-setting performance from changing setting proportions.
- Which labelled step matches this detail? New accuracy 28/50=56%; more use now occurs in the harder dim setting.
- 1. New pooled rate
- 2. Keep slice rates
- 3. Old pooled rate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: New pooled rate New accuracy 28/50=56%; more use now occurs in the harder dim setting.

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

Changed-case question and optional hint

- Changed-case practice: New-use mix returns to40 bright and 10 dim with the same slice rates. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Pooled accuracy returns to74%.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Pooled accuracy returns to74%. The mixture weights change, while each slice's conditional performance stays fixed.

Repair and retain the record

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

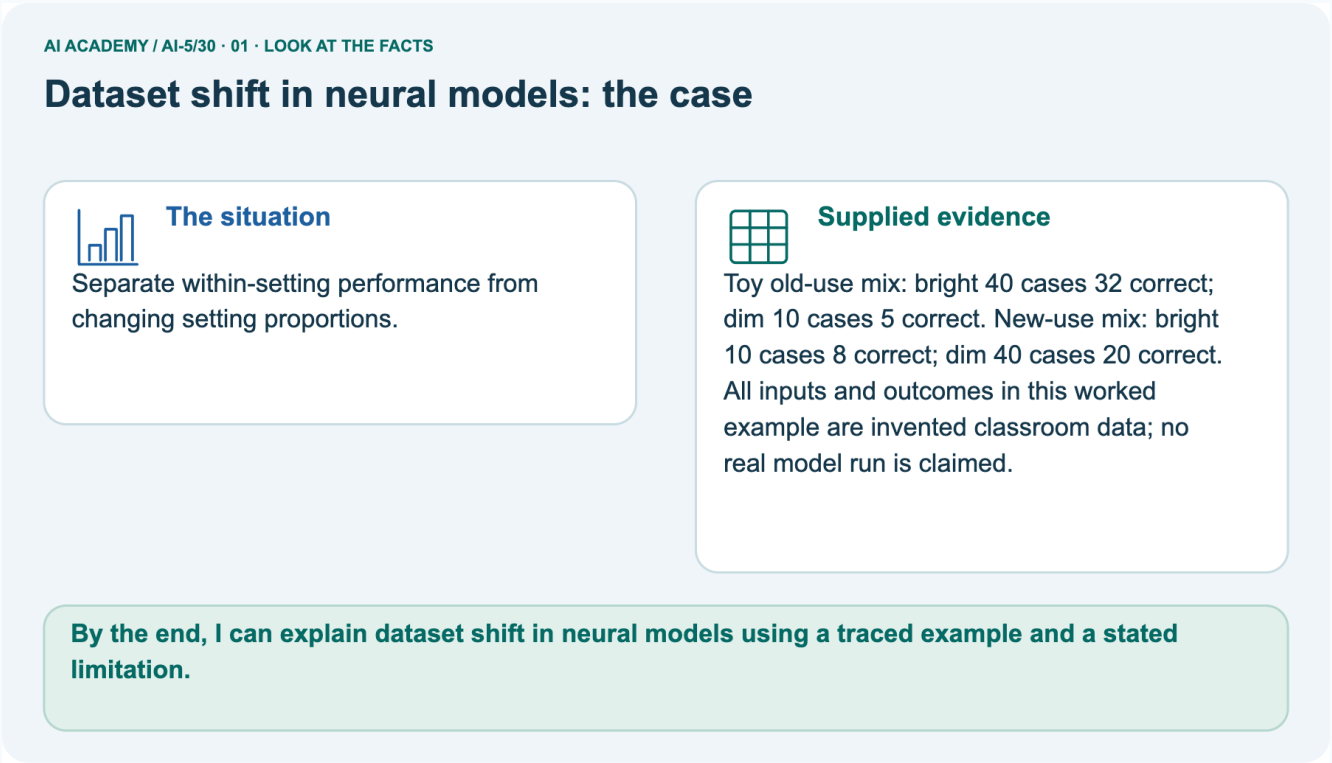
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

Choose the picture that helps explain the current step.

01. Dataset shift in neural models: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

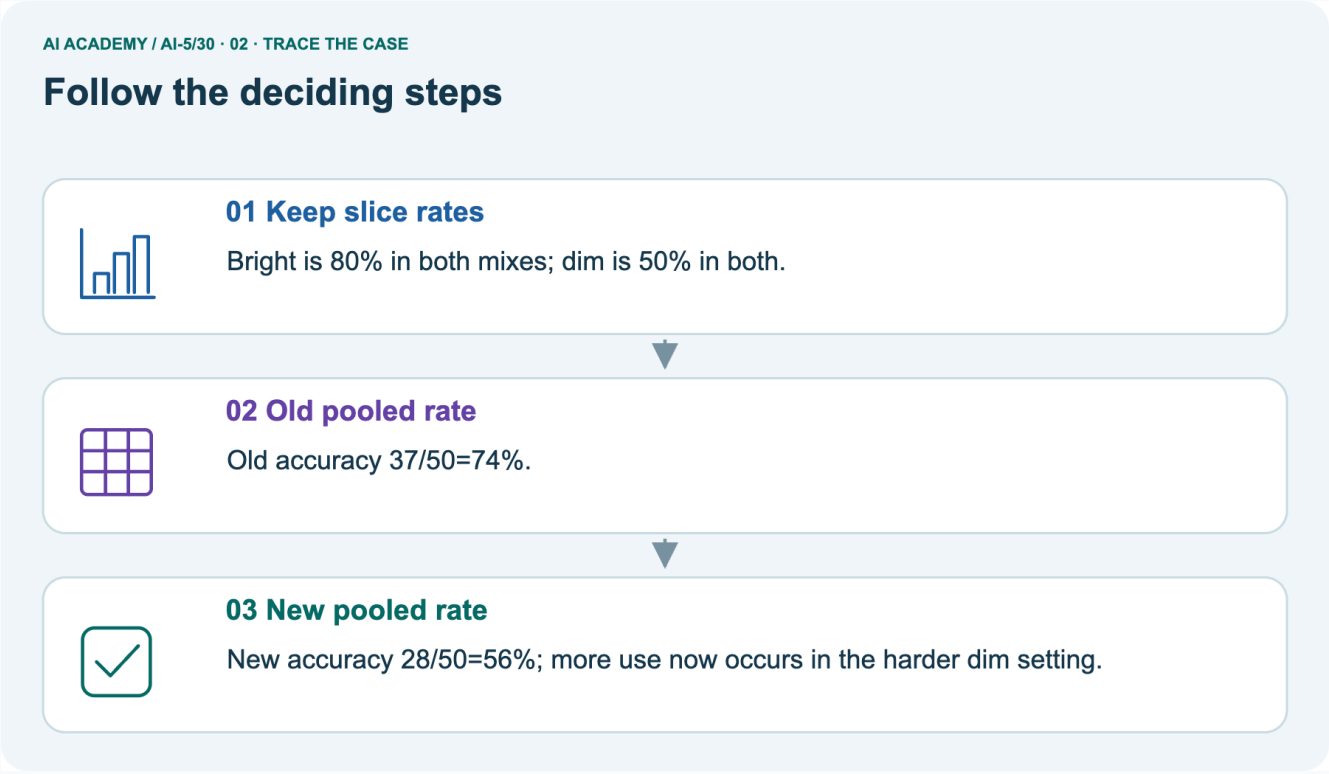
Separate within-setting performance from changing setting proportions. Supplied facts: Toy old-use mix: bright 40 cases 32 correct; dim 10 cases 5 correct. New-use mix: bright 10 cases 8 correct; dim 40 cases 20 correct.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



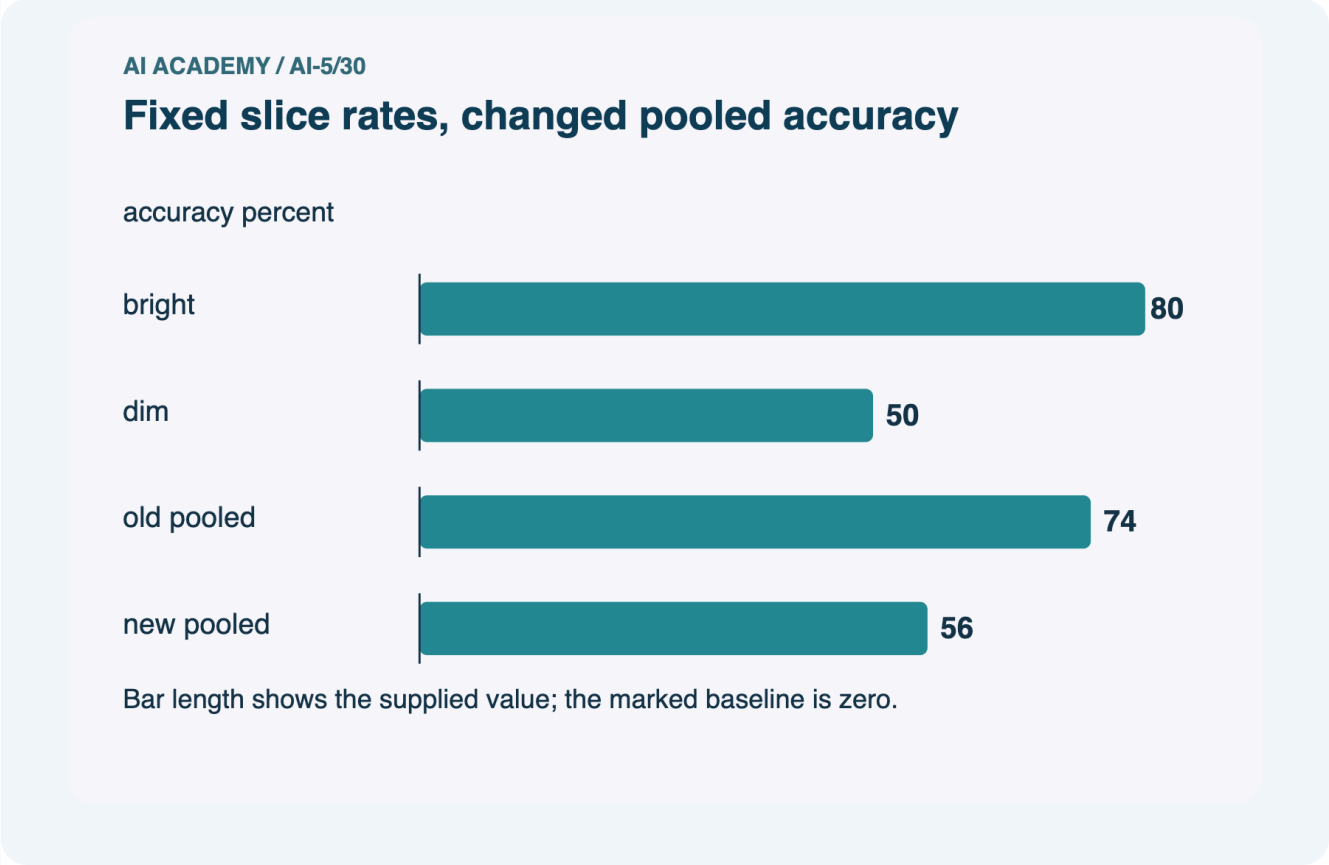
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep slice rates: Bright is **80%** in both mixes; dim is 50% in both. → Old pooled rate: Old accuracy 37/50=74%. → New pooled rate: New accuracy 28/50=56%; more use now occurs in the harder dim setting.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Fixed slice rates, changed pooled accuracy



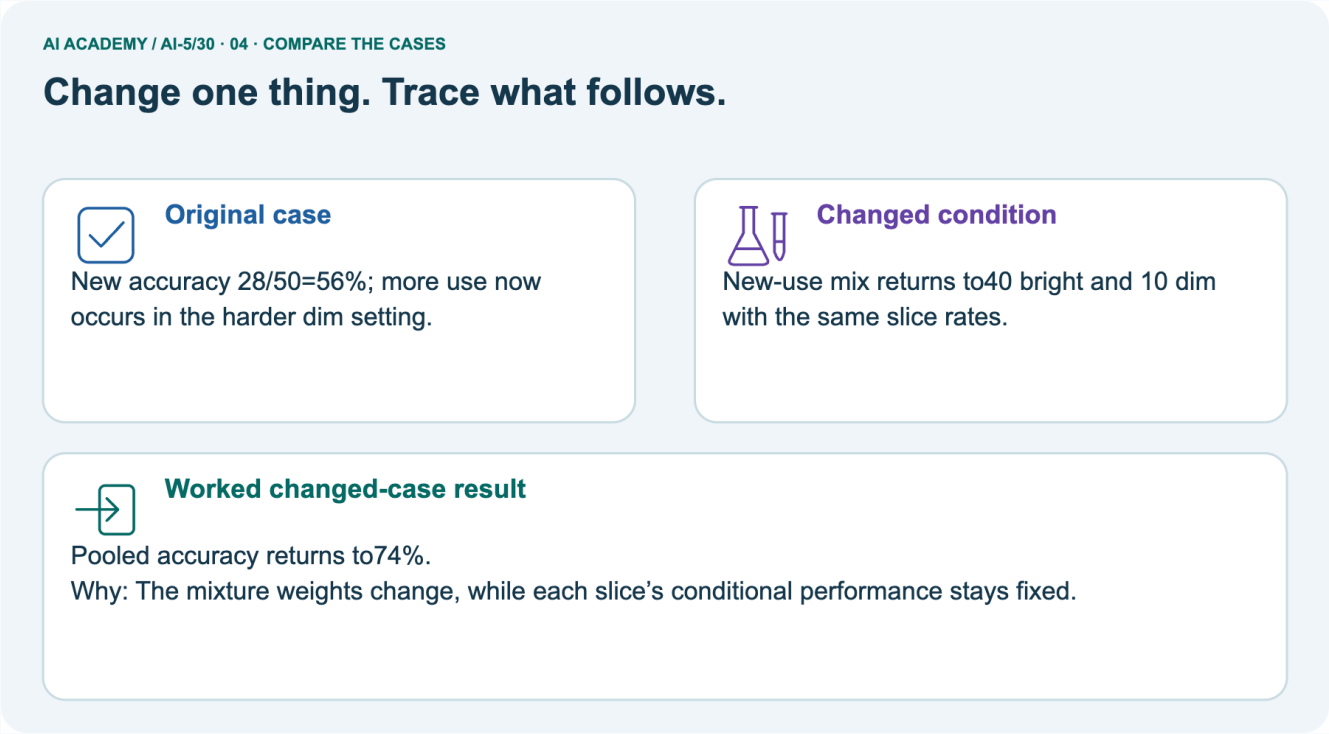
Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed slice rates, changed pooled accuracy. Units: accuracy percent. bright: 80. dim: 50. old pooled: 74. new pooled: 56.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

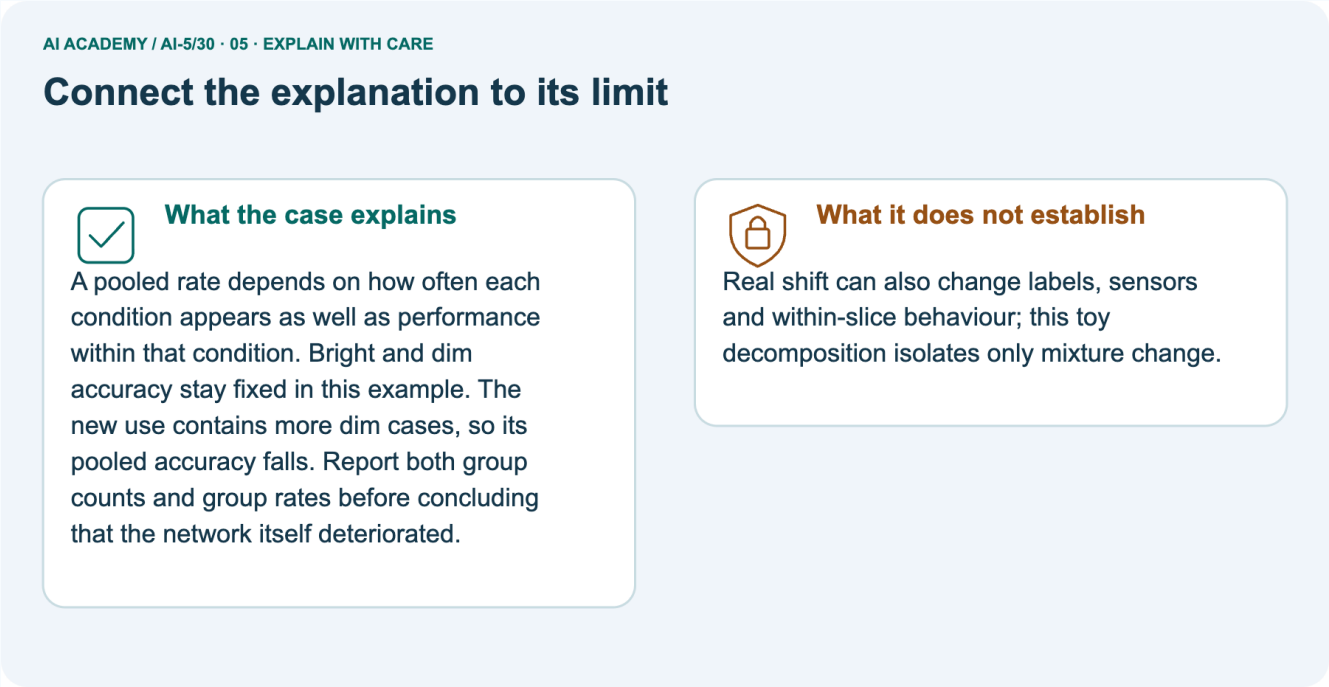
Original case: New accuracy 28/50=56%; more use now occurs in the harder dim setting.
Changed condition: New-use mix returns to40 bright and 10 dim with the same slice rates.
Expected result: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice's conditional performance stays fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

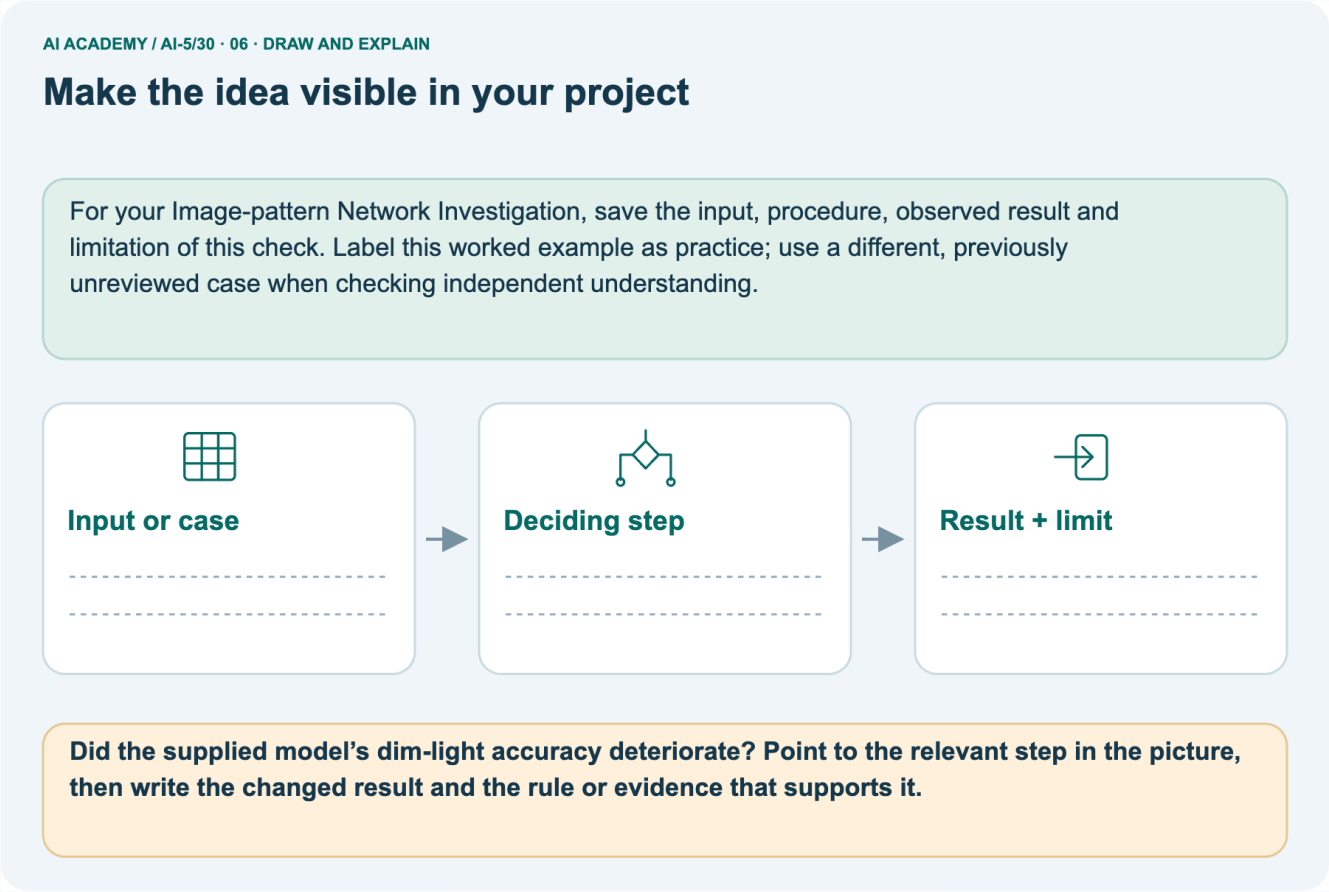
Report both group counts and group rates before concluding that the network itself deteriorated.

Limit: Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Dataset shift in neural models: the case

02. Follow the deciding steps

03. Fixed slice rates, changed pooled accuracy

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.